# Michele Cespa

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## Education:

**PhD Statistics & Machine Learning, University of Oxford** [2026-2030]
*Jesus College, part of StatML CDT*

**MSc Artificial Intelligence, Imperial College** [2025-2026]
*Salter's Fellowship Scholar*

- Relevant Modules: Information Theory, Probabilistic Inference, Non-Euclidean ML, Deep Learning
- Group SWE Project: LLM Uncertainty, Robustness & Security with OLMo (see here)

**BA (Hons) Natural Sciences, University of Cambridge** [2022-2025]
*Girton College*

- Grade: 1st Class, ranking 14th in cohort of 129
- Relevant Modules: Statistical Physics, Electrodynamics, General Relativity, Soft Matter Physics
- Courseworks: Dynamics in Complex Fluids, Phase Locked Loops, Physics Informed Neural Networks
- Activities: member of both the Odyssey Fellowship and Cambridge AI Safety Hub Fellowship

**City of London School** [2015-2022]
*Academic Scholar*

- A-Levels: (all A*) Physics (top of year), Chemistry, Mathematics, Further Mathematics

## Experience:

**Machine Learning Research Intern** [2026]
*MSc Thesis, supervised by Prof. Pedro A.M. Mediano alongside Prelego.ai*

- Worked on ML models for hospital acquired rare disease prognosis
- Conducted research on: foundations of the XGBoost algorithm, unsupervised out-of-distribution adaptation and causal inference for model performance through treatment assignment bias

**Meta Software Engineering Summer Intern** [2025]
*WhatsApp Team*

- Fullstack iOS developer working on saved content in WhatsApp
- Learned C, Objective-C, Swift and Sapling (source control system similar to Git)

**Research Intern, University of Cambridge** [2024]
*ARIA funded lab, Biochemistry Department*

- Built Raspberry Pi operated bioreactors to study organism growth dynamics
- Wrote Python library to operate the reactors (see here)
- Learned about Koopman operator theory & Dynamic Mode Decomposition

**Zero Gravity (charity) Volunteer** [2024]

- Mentored and tutored an A-Level student applying to Natural Sciences (Cambridge)

**Athena Tuition Tutor** [2022]

- Tutored students aged 11-16 in Physics, Maths & Chemistry for 11+, 13+ and As Level exams

## Achievements:

**Salter's Fellowship Scholarship** [2025]
*Imperial College Dept. of Computing & Worshipful Company of Salters (£10,000)*

**Academic Scholarship** [2023-2025]
*Girton College, Cambridge (awarded after final year exams all three years totaling £1,200)*

**City of London School Scholar's Prize** [2020-2022]
*For excellence in A-Levels & GCSEs*

## Technical Skills:

**Languages & Libraries:** Python, C, HTML, PyTorch

**Developer Tools:** Git, GitHub, WandB, Linux

## Misc:

**Hobbies:** Calisthenics, Electric Guitar
**Languages:** *Native:* English, Italian; *Intermediate:* Spanish, French